

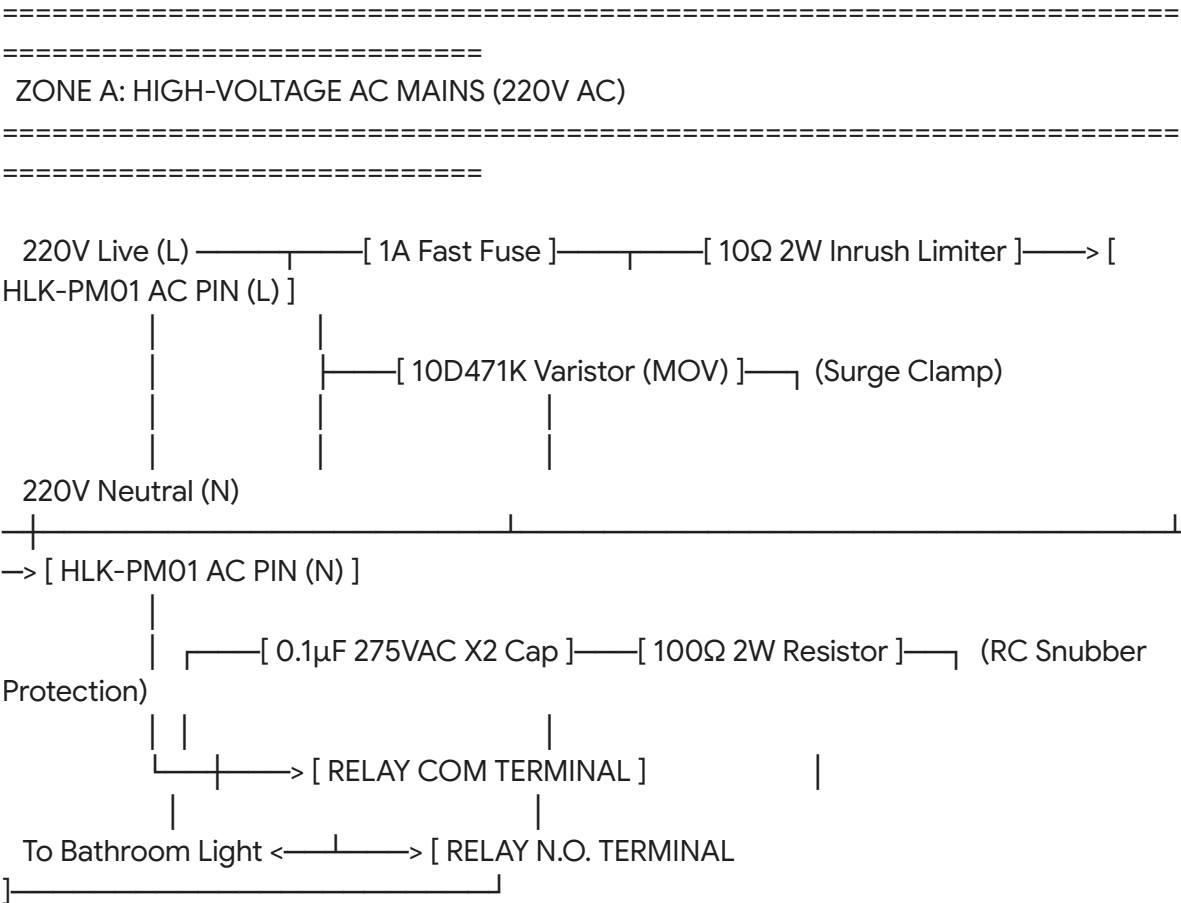
INDUSTRIAL PRODUCTION MANUAL: SMART BATHROOM PRESENCE SENSOR (10–20 YEAR LIFESPAN)

Target Hardware: ESP32-C3 SuperMini | HLK-LD2410B (24GHz FMCW mmWave Radar) | HLK-PM01 (5V 3W Isolated AC-DC)

System Architecture: Dual-Stack Redundancy (Sub-10ms Local Interrupt + LAN MQTT + ESP-NOW Peer-to-Peer Backup)

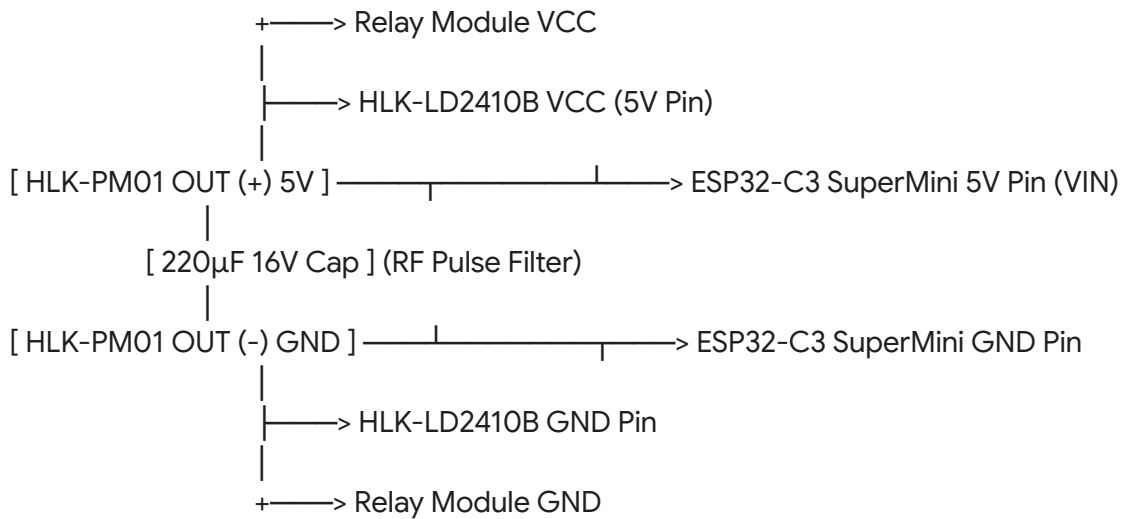
1. Complete System Circuit & Wiring Schematic

To prevent surface arcing in humid bathroom environments, the Zero PCB layout is divided into two distinct physical zones separated by a **2mm air-gap isolation slot**.



[PCB CREEPAGE AIR GAP / 2MM PHYSICAL ISOLATION SLOT CUT THROUGH FIBERGLASS]

ZONE B: LOW-VOLTAGE DC LOGIC (5V / 3.3V)



SIGNAL LOGIC CONNECTIONS:

- * HLK-LD2410B [OUT] —————> ESP32-C3 SuperMini [GPIO 8] (Hardware Interrupt Input)
- * Relay Module [IN] —————> ESP32-C3 SuperMini [GPIO 10] (Relay Actuation Output)

2. Comprehensive Pinout & Interconnection Table

Sub-Assembly	Pin / Terminal	ESP32-C3 SuperMini Pin	Board Track / Power Rail Connection	Engineering Purpose
HLK-PM01	AC Input L	—	Fused 220V	Mains AC input

Power Module			Live Line	(via 1A Fuse & Inrush Resistor).
HLK-PM01 Power Module	AC Input N	—	220V Neutral Line	Mains AC return line.
HLK-PM01 Power Module	DC Output +V0	5V (VIN)	5V DC Power Rail	Regulated low-voltage DC source.
HLK-PM01 Power Module	DC Output -V0	GND	Common Ground Rail	Common system ground.
HLK-LD2410B Radar	VCC	—	5V DC Power Rail	5V power input for 24GHz RF oscillator.
HLK-LD2410B Radar	GND	—	Common Ground Rail	Common ground reference.
HLK-LD2410B Radar	OUT	GPIO 8	Logic Input	Sends digital HIGH on human presence; triggers ISR.
5V Relay Module	VCC	—	5V DC Power Rail	Low-voltage DC supply for relay optocoupler.
5V Relay Module	GND	—	Common Ground Rail	Common ground reference.
5V Relay	IN	GPIO 10	Logic Output	Driven HIGH by ESP32-C3

Module				to actuate mains load.
5V Relay Module	COM	—	Fused 220V Live Line	AC input from grid to relay switch contact.
5V Relay Module	N.O.	—	Bathroom Light (Live)	Normally Open contact; closes to illuminate light.
220μF 16V Capacitor	Positive (+)	—	5V DC Power Rail	Absorbs 24GHz RF current spikes to prevent brownout.
220μF 16V Capacitor	Negative (-)	—	Common Ground Rail	Ground reference for filter capacitor.
RC Snubber Circuit	0.1μF + 100Ω	—	Across Relay COM & N.O.	Suppresses inductive/capacitive arcing across contacts.

3. Step-by-Step Hardware Fabrication Procedure (Zero PCB)

Step 1: Mechanical Board Division & Isolation Slot

1. Mark a clear dividing line across your universal Zero PCB, allocating 40% of the area to **Zone A (High-Voltage AC)** and 60% to **Zone B (Low-Voltage DC)**.
2. Maintain a **minimum 6mm creepage distance** between any copper trace in Zone A and any trace in Zone B.
3. Using a Dremel, rotary cut-off tool, or fine hacksaw blade, cut a **2mm-wide physical slot through the fiberglass board** along the dividing line. This eliminates surface moisture

tracking in humid bathroom air.

Step 2: High-Voltage Section Assembly (Zone A)

1. Solder the **1A Glass Fuse Holder** at the immediate 220V Live input terminal.
2. Place the **10D471K Varistor (MOV)** in parallel across the Live (post-fuse) and Neutral tracks.
3. Solder the **10Ω 2W Flameproof Resistor** in series between the fuse output and the HLK-PM01 AC-L pin.
4. Mount the **HLK-PM01** power supply module and the **5V Opto-Isolated Relay Module**.
5. Assemble the **RC Snubber** by soldering the **0.1μF 275VAC X2 safety capacitor** and **100Ω 2W resistor** in series directly across the relay's COM and N.O. pins.
6. **Reinforce AC Traces:** Strip a strand of 1.5mm copper household wire and solder it along the bottom copper tracks of the 220V Live and Neutral lines to handle peak switching loads without trace overheating.

Step 3: Low-Voltage Section Assembly (Zone B)

1. Solder **female header sockets** for the ESP32-C3 SuperMini to allow easy replacement.
2. Solder the **220μF 16V electrolytic capacitor** directly across the 5V and GND output traces of the HLK-PM01 to act as an RF current reservoir.
3. Remove the female DuPont connectors from the **SH1.27mm 5-Pin cable**, strip 3mm of insulation from the VCC, GND, and OUT wires, and solder them directly into the Zero PCB tracks. *(Do not use DuPont spring contacts for permanent 10–20 year installations).*

Step 4: Moistureproofing & Enclosure Mounting

1. Clean the completed solder side of the Zero PCB using Isopropyl Alcohol (IPA) and a stiff brush to remove residual flux.
2. Apply **2 to 3 even coats of clear Acrylic Conformal Coating** over the soldered PCB surfaces (both top and bottom). **Critical Warning:** Never apply conformal coating to the ESP32-C3 antenna, the LD2410B radar antenna face, or the glass fuse cartridge.
3. Mount the board inside an **IP65/IP67 ABS Plastic Waterproof Junction Box** located high on the bathroom wall, orienting the LD2410B sensor so it faces toward the toilet/shower area through the plastic casing.

4. HLKRadarTool Bluetooth Calibration Protocol

To prevent false triggers from hallways while detecting micro-motion (e.g., chest breathing while sitting still), configure the LD2410B distance gates as follows:

[Gate 0: Threshold] —> [Gate 1: Vanity/Sink] —> [Gate 2: Toilet Area] —> [Gate 3: Wall Boundary]

Move: 60 | Stat: 40 Move: 60 | Stat: 40 Move: 50 | Stat: 15 Move: 0 | Stat: 0

1. **Set Boundary Limit (Max Gates):** Measure the physical distance from your sensor to the opposite bathroom wall. Set both **Max Moving Gate** and **Max Static Gate** to **3** (covering $3 \times 0.75 \text{ m} = 2.25 \text{ m}$). This blinds the sensor to movement happening outside the bathroom.
2. **Configure Gate Sensitivities:**
 - **Gate 0 & 1 (Doorway & Vanity):** Set Moving Sensitivity = **60**, Static Sensitivity = **40**.
 - **Gate 2 (Toilet / Primary Still Zone):** Set Moving Sensitivity = **50**, Static Sensitivity = **15** (*lower value = higher sensitivity to breathing*).
 - **Gates 3 through 8 (Outside Zone):** Set Moving Sensitivity = **0**, Static Sensitivity = **0**.
3. **Set Unmanned Duration:** Set **Unmanned Duration** to **5 seconds** in the app. This creates a hardware-level debounce window before the OUT pin goes LOW.

5. Enterprise Production Firmware (ESP32-C3 SuperMini)

This non-blocking C++ firmware utilizes hardware interrupts for instantaneous sub-10ms light actuation, OTA update support, a watchdog timer, and dual-stack MQTT/ESP-NOW network reporting.

```
#include <WiFi.h>
#include <PubSubClient.h>
#include <esp_now.h>
#include <ArduinoOTA.h>
#include <esp_task_wdt.h>

// --- PIN ASSIGNMENTS ---
#define RADAR_OUT_PIN      8 // HLK-LD2410B Digital Output
#define RELAY_PIN          10 // Relay Actuation Line
#define WATCHDOG_TIMEOUT_SEC 5

// --- NETWORK CREDENTIALS ---
const char* WIFI_SSID     = "YOUR_WIFI_SSID";
const char* WIFI_PASSWORD = "YOUR_WIFI_PASSWORD";
const char* MQTT_SERVER   = "192.168.1.100"; // Home Assistant Broker IP
const int  MQTT_PORT      = 1883;

// Master ESP32-WROOM MAC Address for ESP-NOW Peer-to-Peer Backup
uint8_t masterMacAddress[] = {0x24, 0x0A, 0xC4, 0xAA, 0xBB, 0xCC};
```

```

// --- OBJECT INSTANCES ---
WiFiClient espClient;
        mqttClient        ;

// --- VOLATILE & TIMING VARIABLES ---
volatile bool presenceDetected = false;
bool lastRelayState = false;
unsigned long lastPresenceTimestamp = 0;
const unsigned long OFF_TIMEOUT_MS = 3000; // 3-Second Local Failsafe Off-Delay
unsigned long lastNetworkRetry = 0;

typedef struct struct_message {
    char nodeID[16];
    bool lightActive;
} struct_message;
struct_message telemetryPacket;

// --- HARDWARE INTERRUPT SERVICE ROUTINE (ISR) ---
void        handleRadarInterrupt {
    presenceDetected = (digitalRead(RADAR_OUT_PIN) == HIGH);
}

// --- TELEMETRY BROADCAST FUNCTION ---
void transmitStatus bool {
    telemetryPacket.lightActive = state;

    // 1. ESP-NOW Redundant Peer Broadcast
    esp_now_send(masterMacAddress, (uint8_t *)&telemetryPacket,
sizeof(telemetryPacket));

    // 2. LAN MQTT Topic Publication
    if (mqttClient.connected()) {
        mqttClient.publish("home/bathroom/light/state", state ? "ON" : "OFF", true);
    }
}

void setup {
    Serial.begin(115200);

    // Initialize Actuation & Sensing Hardware
    pinMode(RELAY_PIN, OUTPUT);
    digitalWrite(RELAY_PIN, LOW); // Failsafe Startup State: Light OFF

```

```

pinMode(RADAR_OUT_PIN, INPUT_PULLDOWN);

// Attach Fast Hardware Interrupt (<10ms Response)
attachInterrupt(digitalPinToInterrupt(RADAR_OUT_PIN), handleRadarInterrupt,
CHANGE);

// Configure Task Watchdog Timer
esp_task_wdt_init(WATCHDOG_TIMEOUT_SEC, true);
esp_task_wdt_add(NULL);

// Initialize Wi-Fi Layer
WiFi.mode(WIFI_STA);
WiFi.begin(WIFI_SSID, WIFI_PASSWORD);

// Initialize ESP-NOW Protocol
if (esp_now_init() == ESP_OK) {
    esp_now_peer_info_t peerInfo = {};
    memcpy(peerInfo.peer_addr, masterMacAddress, 6);
    peerInfo.channel = 0;
    peerInfo.encrypt = false;
    esp_now_add_peer(&peerInfo);
}

// Configure MQTT & OTA
mqttClient.setServer(MQTT_SERVER, MQTT_PORT);
strcpy(telemetryPacket.nodeID, "BATH_C3_NODE");
ArduinoOTA.setHostname("Bathroom-Presence-Node");
ArduinoOTA.begin();

// Verify Initial Radar State on Boot
presenceDetected = (digitalRead(RADAR_OUT_PIN) == HIGH);
}

void loop {
    // Reset Watchdog Timer on Every Cycle
    esp_task_wdt_reset();

    // --- STEP 1: LOCAL HARDWARE EXECUTION (Zero Network Latency) ---
    bool currentPresence = presenceDetected;

    if (currentPresence) {
        lastPresenceTimestamp = millis(); // Refresh presence expiration timer
        if (!lastRelayState) {

```



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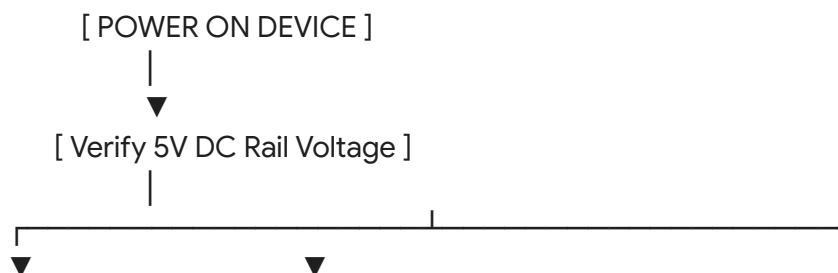
    digitalWrite(RELAY_PIN, HIGH); // ACTUATE LIGHT INSTANTLY
    lastRelayState = true;
    transmitStatus(true);
  }
} else {
  // Enforce 3-Second Hard Timeout After Radar Drops
  if (lastRelayState && (millis() - lastPresenceTimestamp >= OFF_TIMEOUT_MS)) {
    digitalWrite(RELAY_PIN, LOW); // DE-ACTUATE LIGHT
    lastRelayState = false;
    transmitStatus(false);
  }
}

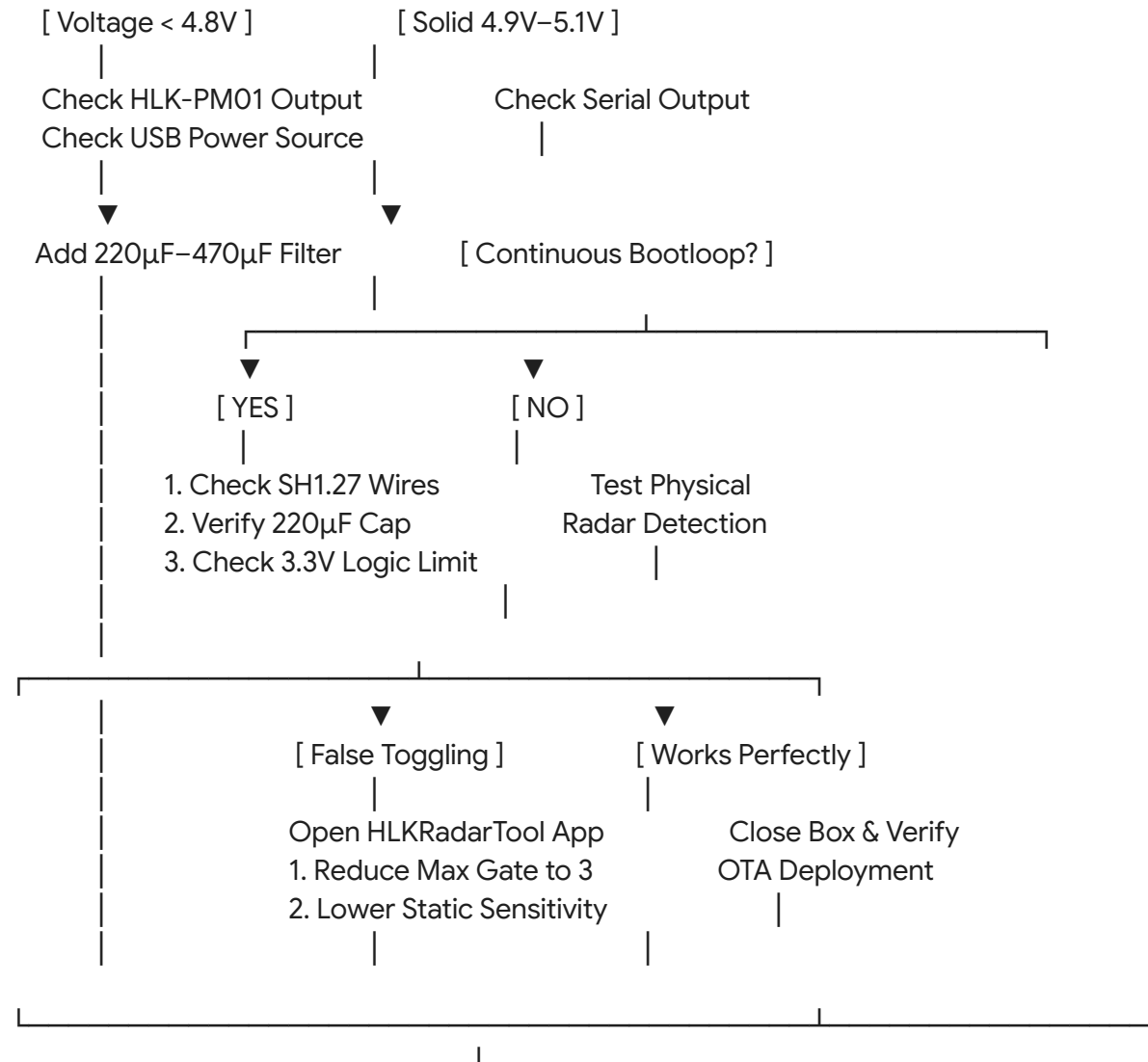
// --- STEP 2: ASYNCHRONOUS NETWORK & OTA MAINTENANCE ---
if (WiFi.status() == WL_CONNECTED) {
  ArduinoOTA.handle();

  if (!mqttClient.connected()) {
    if (millis() - lastNetworkRetry > 5000) { // Limit retry flood to 5s intervals
      lastNetworkRetry = millis();
      if (mqttClient.connect("ESP32_Bathroom_Node")) {
        transmitStatus(lastRelayState);
      }
    }
  } else {
    mqttClient.loop();
  }
}
}

```

6. Complete Debugging & Fine-Tuning Protocol





Hardware Troubleshooting Checklist

- **Symptom: ESP32-C3 SuperMini Bootloops (SPI_FAST_FLASH_BOOT / POWERON_RESET)**
 - **Cause:** Voltage sag on the VIN (5V) line caused by thin wiring or sudden RF current spikes when the LD2410B oscillator fires.
 - **Fix:** Ensure the **220µF 16V capacitor** is soldered directly across the 5V and GND pins on the Zero PCB. Solder SH1.27mm wires directly to the board tracks instead of using DuPont breadboard headers.
- **Symptom: Relay Clicks Continuously or Arcs with LED Bulb**
 - **Cause:** High capacitive inrush current from the LED bulb's internal driver ballasts.
 - **Fix:** Verify the **0.1µF 275VAC X2 safety film capacitor** and **100Ω 2W resistor** are properly wired in series across the relay's COM and N.O. terminals.
- **Symptom: Light Turns ON When Someone Walks in the Hallway**

- **Cause:** Radar microwave penetration through open doors or thin gypsum/brick walls.
- **Fix:** In the **HLKRadarTool** app, reduce **Max Moving Gate** and **Max Static Gate** to **3** (2.25m maximum depth).
- **Symptom: Light Turns OFF While Sitting Still on the Toilet**
 - **Cause:** Static sensitivity threshold is too high to detect micro-motion breathing.
 - **Fix:** In the **HLKRadarTool** app, change Gate 2 **Static Sensitivity** from default down to **15** or **20**. Ensure **Unmanned Duration** is set to **5 seconds**.
- **Symptom: Wireless Code Update Fails After Box Sealing**
 - **Cause:** Incorrect hostname or firewall blocking local OTA mDNS ports.
 - **Fix:** In Arduino IDE, go to **Tools > Port > Network ports** and select **Bathroom-Presence-Node at 192.168.x.x**. Verify your router allows peer-to-peer Wi-Fi communication across local devices.

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